Q-3. (i) $(x^2D^2+1)y = 3x^2$ where $D = \frac{d}{dx}$

Let $x = e^z$ or $z = \log x$

$$\Rightarrow \frac{dz}{dx} = \frac{1}{x}$$

Now $$\frac{dy}{dx} = \frac{dy}{dz}\cdot\frac{dz}{dx} = \frac{1}{x}\frac{dy}{dz}$$

$$\Rightarrow x\frac{dy}{dx} = \frac{dy}{dz} \quad \text{or} \quad xD = D_1 \quad \text{where } D_1 = \frac{d}{dz}$$

Now $$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d}{dx}\left(\frac{1}{x}\frac{dy}{dz}\right) = -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\frac{d}{dx}\left(\frac{dy}{dz}\right)$$

$$= -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x}\frac{d}{dz}\left(\frac{dy}{dz}\right)\cdot\frac{dz}{dx}$$

$$= -\frac{1}{x^2}\frac{dy}{dz} + \frac{1}{x^2}\frac{d^2y}{dz^2}$$

$$\Rightarrow x^2\frac{d^2y}{dx^2} = \frac{d^2y}{dx} - \frac{dy}{dz}$$

$$\text{or} \quad x^2D^2y = (D_1^2 - D_1)y = D_1(D_1-1)y$$

Similarly, we can generalize -

$$\boxed{x^n D^n = D_1(D_1-1)\dots(D_1-n+1)}$$

Now, in given question -

$$(x^2D^2+1)y = (D_1(D_1-1)+1)y = 3e^{2z}$$

$$\text{or} \quad (D_1^2 - D_1 + 1)y = 3e^{2z}$$

The auxiliary eqn. is - $D_1^2 - D_1 + 1 = 0$

After solving $D_1 = (1 \pm i\sqrt{3})/2$

So, $$C.F = e^{z/2}\left[c_1\cos(z\sqrt{3}/2) + c_2\sin(z\sqrt{3}/2)\right]$$

or

$$CF = x^{1/2}\left[c_1\cos\{(\sqrt{3}/2)\log x\} + c_2\sin\{(\sqrt{3}/2)\log x\}\right]$$

as $x = e^z$

and $$P.I = \frac{1}{D_1^2 - D_1 + 1}\, 3e^{2z} = 3\,\frac{1}{2^2 - 2 + 1}\, e^{2z} = e^{2z} = x^2$$

Hence, the required solution –

$$y = C.F + P.I$$

$$y(x) = x^{1/2}\left[c_1\cos\{(\sqrt{3}/2)\log x\} + c_2\sin\{(\sqrt{3}/2)\log x\}\right] + x^2$$

Ans

(ii) $(xD^3 + D^2)y = 1/x \;=\; (x^2D^3 + xD^2)y = 1$

or $(x^3D^3 + x^2D^2)y = x$

After using $x = e^z$, $x^2D^2 = D_1(D_1 - 1)$

$x^3D^3 = D_1(D_1-1)(D_1-2)$

(as (i))

Given eqn becomes –

$$(D_1^3 - 2D_1^2 + D_1)y = e^z$$

C.F. can be calculated as above –

$$P.I = \frac{1}{D_1^3 - 2D_1^2 + D_1}\, e^z = \frac{1}{(D_1-1)^2}\cdot\frac{1}{D_1}e^z = \frac{1}{(D_1-1)^2}\, e^z \quad \left(\because \frac{1}{D_1}e^z = \int e^z dz = e^z\right)$$

$$= \frac{z^2}{2!}e^z \quad \text{since } \frac{1}{(D_1-\alpha)^m}e^{\alpha z} = \frac{z^m}{m!}e^{\alpha z}$$

$$= \left(\frac{x}{2}\right)(\log x)^2$$

And required solution is –

$$y(x) = CF + PI$$

(iii) $(x^2D^2 - 3xD + 4)y = 2x^2$

Similarly –

$C.F = (C_1 + C_2 z)e^{2z} = (C_1 + C_2 \log x)x^2$

$P.I = z^2 e^{2z} = (\log x)^2 x^2$

And $y(x) = CF + P.I$

(iv) $(x^2D^2 - 2)y = x^2 + (1/x)$

It will become – $(D_1^2 - D_1 - 2)y = e^{2z} + e^{-z}$

$\therefore CF = c_1 e^{2z} + c_2 e^{-z} = c_1 x^2 + c_2 x^{-1}$

$P.I = \frac{1}{(D_1^2 - D_1 - 2)}(e^{2z} + e^{-z}) = \frac{1}{(D_1-2)(D_1+1)} e^{2z} + \frac{1}{(D_1+1)(D_1-2)} e^{-z}$

After solving $PI = \frac{1}{3} \log x \,(x^2 + x^{-1})$.

And $y(x) = CF + P.I$

(v) $(x^2D^2 - 3xD + 5)y = \sin(\log x)$

Same as above.

Use for P.I

$\frac{1}{f(D)} \sin mx = \frac{1}{f(-m)} \sin mx$

Soln. is –

$y(x) = x^2 [c_1 \cos(\log x) + c_2 \sin(\log x)] + \frac{1}{8} \times [\sin(\log x) + \cos(\log x)]$.

(vi) $(3x^2D^2 - 5xD + 5)y = \sin(\log x)$

Similarly $y(x) = c_1 x + c_2 x^{5/3} + \frac{1}{16}[\sin(\log x) + \cos(\log x)]$

(vii) Ans→ $y(x) = c_1 x^{-2} + x\,[c_2 \cos(\sqrt{3}\log x) + c_3 \sin(\sqrt{3}\log x)] + 8\cos(\log x) - \sin(\log x)$.

(viii) $(x^4 D^4 + 6x^3 D^3 + 4x^2 D^2 - 2xD - 4)y = 2\cos(\log x)$.

After putting $x = e^z$ or $z = \log x$

and $D_1 = \frac{d}{dz}$

After solving—

we get — $(D_1^4 - 3D_1^2 - 4)y = 2\cos z$

And, final answer —

$y(x) = c_1 x^2 + c_2 x^{-2} + c_3 \cos(\log x) + c_4 \sin(\log x) - \frac{1}{5}\log x \cdot \sin(\log x)$

Ans

Q-4. Given —

$$\frac{dy}{dx} = 2y - 2x^2 - 3 \quad , \quad y(0) = 2$$

We know that the nth approximation y_n of initial value problem —

$$\frac{dy}{dx} = f(x,y) \quad , \quad \text{with } y(x_0) = y_0$$

is given by —

$$y_n = y_0 + \int_{x_0}^{x} f(x, y_{n-1})\, dx$$

Here $f(x,y) = 2y - 2x^2 - 3$, $x_0 = 2$, $y_0 = 2$

$$\therefore \quad y_n = 2 + \int_0^x (2y_{n-1} - 2x^2 - 3)\, dx$$

First approximation —

$$y_1 = 2 + \int_0^x (2y_0 - 2x^2 - 3)\, dx = 2 + \int_0^x (1 - 2x^2)\, dx$$

$$= 2 + x - \frac{2x^3}{3}$$

Second approximation —

$$y_2 = 2 + \int_0^x (2y_1 - 2x^2 - 3)\, dx$$

After solving —

$$y_2 = 2 + x + x^2 - \frac{2x^3}{3} - \frac{x^4}{3}$$

Third approximation :-

$$y_3 = 2 + \int_0^x (2y_2 - 2x^2 - 3)\,dx$$

After solving -

$$y_3 = 2 + x + x^2 - \frac{x^4}{3} - \frac{2x^5}{15}$$

So $$\boxed{y(x) = 2 + x + x^2 - \frac{x^4}{3} - \frac{2x^5}{15}}$$

Q-5 Given - $\frac{dy}{dx} = x + y^2$, $y(0) = 0$

Similarly, as 4th -

First approximation - $y_1 = \frac{1}{2}x^2$

Second approximation - $y_2 = \frac{1}{2}x^2 + \frac{1}{20}x^5$

Third approximation - $y_3 = \frac{1}{2}x^2 + \frac{1}{20}x^5 + \frac{1}{4400}x^{11} + \frac{1}{160}x^8$.

So $$\boxed{y(x) = \frac{1}{2}x^2 + \frac{1}{20}x^5 + \frac{1}{4400}x^{11} + \frac{1}{160}x^8}$$